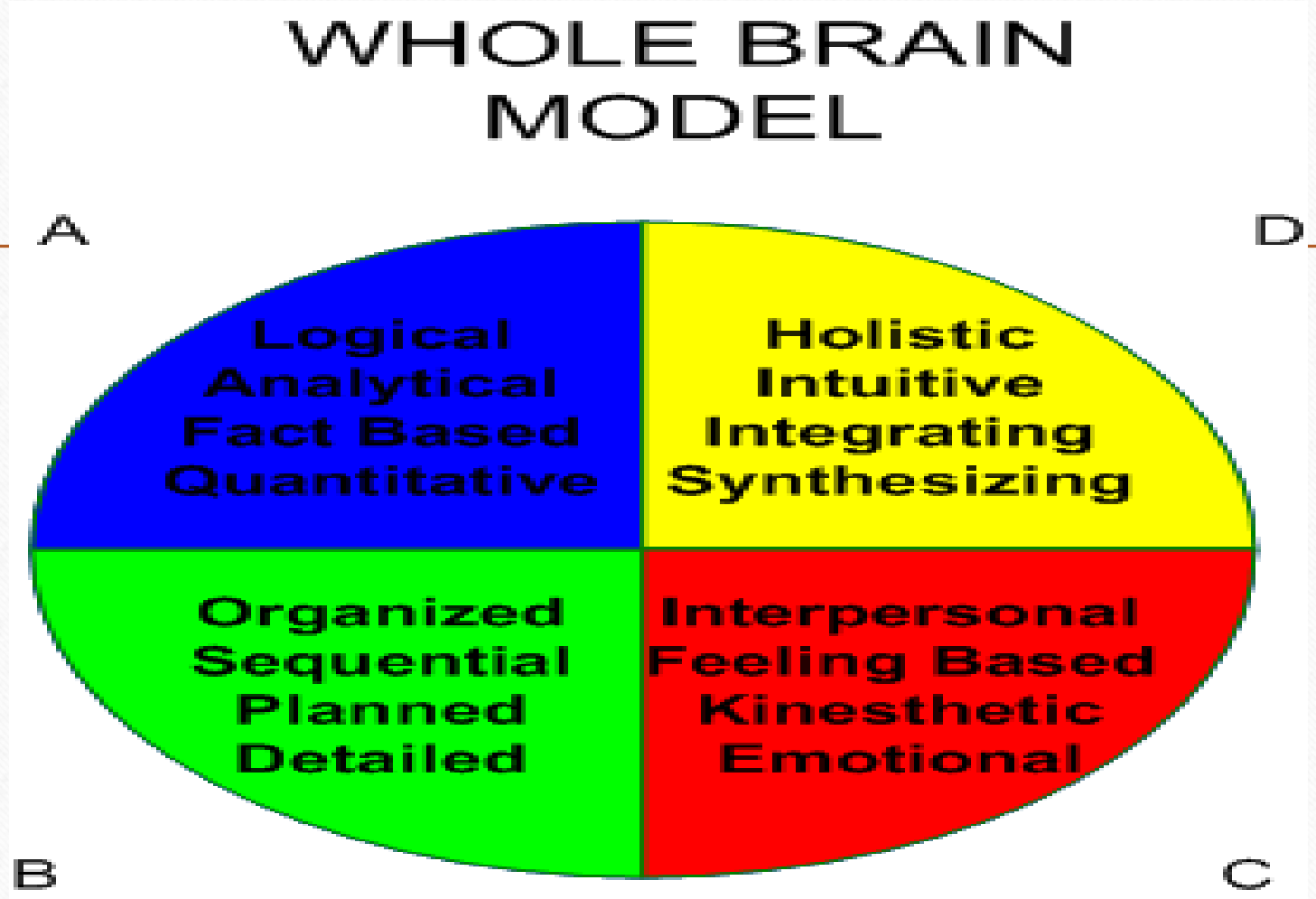
Week 5

Creativity Theories and Thinking

Theories of Creative Thinking and the Creative Problem Solving Process

- Whole Brain Model
- Walla's model of the creative process
- Divergent and Convergent thinking
- Conditions of creative thinking
- Theories of creative problem solving
- Creative problem solving

WHOLE BRAIN MODEL



Whole Brain model contd

- Herrmann ‘s model is based on Learning/ thinking /doing
- A and B sides are the preparation and verification
- C and D sides of the brain are the incubation and illumination
- This model looks at how you first look at a problem it creates a thinking profile of you and how you respond to that quadrant
- Preferred mode is 1 and Competent mode is 2 (this is the way you function but may not be your 1st choice) and Avoidance is given a 3. You may have more than 1 quadrant with the same number

Walla's Model of Creative Thinking

1. Preparation – focuses on the problem and explores the problem's dimensions
2. Incubation – Subconscious mulling of the problem
3. Intimation – Inkling that a solution may well be on its way
4. Illumination – Discovery "eureka", the idea bursts through…
5. Verification – Focus on the practicality, effectiveness and appropriateness

It is a structured approach to creative problem solving – it needs time and a variety of people

Major theories of Creativity

- **Pyschoanalytical Theory of Creativity** – Jung; Freud, Adler and Hammer
- Main tenet is people become more creative as a reaction to difficult circumstances and/or repressed emotions
- When a situation becomes difficult or go through a traumatic event people will pull back from their surroundings. They then rely on their creative side to find a solution to the difficult situation as an outlet for their now repressed emotions
- A sense of inferiority can contribute to creativity and thus the creativity can actually make them feel superior and move forward
- Freud suggest that "unsatisfied wishes are the driving power behind fantasies"

Mental Illness Theory of Creativity

- Briggs; Eisemann; Goodman are proponents of this theory
- For people to become creative, some form of mental illness needs to be present. Doesn't have to be severe, as this can actually impede.
- Common disorders associated are bipolar, manic-depressive, and sever mood swings
- Verhaeghen and Colleagues in a recent study show when a person is in a reflective mode, they become more creative, but can also lead to depression – this is indicative of artists and writers. It is suggested this is when people become sensitive to colours, sounds and emotions
- It is said that stress actually impairs creativity by distracting us…

Addiction Theory of Creativity

- Lapp, Collins, Norlander and Wallas
- The addiction to substances such as alcohol and drugs contributes to creativity and may even cause it in some people
- One tenet of this theory suggests the use of recreational drugs and alcohol removes the stress element that inhibits our creative ability.
- While under the influence, many people may tend to have a train of thought that others would view as loose and unpredictable, but these are traits we relate to creativity
- Arguers against this theory is that addiction doesn't cause the creativity – but those with creative tendencies are more prone to addiction

Creative Theory of Psychoticism

- Eysenck
- The tenet is that all creative people have a disposition for psychotic tendencies. It is said these psychotic tendencies form the basis for creative personalities
- Psychoticism is seen as being halfway between a "normal" person and the state of psychosis
- A psychotic generally rejects social, cultural or authoritarian norms – it is an anti convention form of rebellion
- They are risk takers, impulsive and have a disregard for common sense and spontaneous in nature

Humanistic Theory of Creativity

- Maslow; Rogers and Fromm
- So looking at Maslow's Hierarchy of Needs, once we get to Self Actualisation, we are now free and comfortable enough to express ourselves in a creative manner
- This theory argues that environment is not a factor in creativity – as a person who is able to meet all their needs, they can then choose to be creative
- Creativity is central to our growth and learning processes


Self-actualization
desire to become the most that one can be
Esteem
respect, self-esteem, status, recognition, strength, freedom
Love and belonging
friendship, intimacy, family, sense of connection
Safety needs
personal security, employment, resources, health, property
Physiological needs
air, water, food, shelter, sleep, clothing, reproduction

Maslow's 3 types of Creativity

- **Primary:** Creativity that allows us to escape the daily grind/ stress of life. Thus we channel it into painting, drawing, writing etc
- **Secondary:** A higher level of thought to achieve. It tends to be more thought out and involved than the primary creativity.
- **Integrated:** Combines both. It may be spontaneous when we commence the creative work, a lot more thought goes into what we hope to acjieve before we begin. Eg philosophy and scientific discoveries

Theories of Creative Grace/ Accident

Creativity BY Grace:

- This is the idea of genius , they are endowed with a superhuman potential
- It is seen as a divine gift

Creativity BY Accident:

- It rises by chance
- Accidental discoveries eg Immunisation arising from an interruption in work, radioactivity from a wrong hypothesis

Theories of Creative Association/ Cognitive

Creativity Association:

- Suggests applying procedures from one area to another
- Underlies the justification for many divergent thinking techniques such as lateral thinking and brain storming

Creativity Cognitive:

- It is a normal human activity
- It is about reasoning, recognition and understanding

Divergent and Convergent Thinking

- Divergent thinking is about coming up with many possible solutions to the one problem
- Convergent thinking is solving a problem with only one single correct solution

Conditions of Creative Thinking

- Receptivity
- Immersion
- Seeing questions
- Utilisation of errors
- Detached devotion

Creative Problem Solving

- Objective finding – defines the problem area
- Fact Finding – gather the information
- Problem Finding - defines the problem correctly
- Idea Finding – generates many solutions to the problem
- Solution Finding – evaluates and choose between possible solutions
- Acceptance Finding – implements the chosen ideas correctly

Objective Finding

- Essentially involves Divergent Thinking to generate a list of problems
- Convergence is then used to identify the most relevant problem areas for further exploration
- Hits and hotspots are identified by questioning

1. Ownership – is one motivated to solve it
2. Priority – how important is the problem
3. Critical Nature – how urgent is it to be solved

Fact Finding

- Overall comprehension of the problem is increased by collection of relevant information
- This also helps new ideas to be generated
- Hits and hotspots can assist convergence
- The previously identified problems may now be seen from a new perspective

Problem Finding:

- Essentially uses the previous stage “hits” to identify the most productive problem definition possible

Idea Finding:

- Helps to structure the search for potential solutions
- Mainly divergent activity is used to generate many ideas using a variety of idea generation aids

Solution Finding:

- Is basically the choice of ideas that can be transformed into workable solutions

Acceptance Finding

- Is a divergent activity that helps to implement solutions successfully such as:

1. Listing potential implementation obstacles and ways to overcome them
2. Developing both preventive actions and contingency plans
3. Generating an action plan to implement a solution

Creative Problem Solving cont'd

- Creative problem solving is intuitive and natural. It is not designed to replace our natural creative processes, rather build upon them
- It is also research based, much research has been undertaken on creative problem solving to try and come up with ONE best way
- Creative problem solving can also be very generalizable. It can be used in all aspects of life, other models are a bit more domain specific
- It requires the use of different mental operations, thus it is a higher order thinking skill, that has sub skills to it.

Four preferences to thinking creatively

- There are four ways to have a preference for a creative way of thinking.

1. **The Clarifier:** is the person who says are we really solving the right problem – they want to be quick to move to the solution – they like doing research and gathering information. They attend to the detail, so they are methodical – they tend to ask lots of questions because they crave information
2. **The Ideator:** looks at the big picture. They dream of possibilities. They like to toy with ideas – they stretch their imagination / out of the box ideas, they jump from A- Z. They are adventurous and dreamers

Four preferences to thinking creatively
cont'd

3. **The Developer:** Takes an idea and tinkers with it. They improve upon other's ideas. They put together workable solutions. They analyse competing solutions, looking at both the pros and cons. These thinkers are perfectionists, they can get stuck in developing the perfect solution and other see them as being nit pickers and finding flaws with others solutions

4. The Implementer: They are individuals who are quick to action, always moving forward. They need to see ideas to hit fruition – they want things to happen – they are decisive and determined. Unfortunately they can leap into action too soon, they may appear too pushy and try to oversell their idea.

- Understanding your preference can improve your creative performance.
- By working in teams, you may well have others whom think differently to you and this should lead to complementary ideas.
- It should lead to helping to solve a problem more effectively if we take into account the different paths to the solution/s.
- https://www.youtube.com/watch?v=aH2ll5bwpKw